

DEEP RESEARCH

Regime-Switching Engines

Should TradePilot build 3-4 specialist engines and route between them by market regime? A 5-agent parallel research dive.

5

Parallel research agents

175

LONG signals v5
blocked on 04-29

3 wk

Recommended fix vs 9-
15 wk hard switch

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Companion to PRODUCTION_ROADMAP_v6.1

Verdict (read this first)

You asked: "Build 3-4 specialised engines (BULL/BEAR/SIDEWAYS), one per regime, and a router that switches between them mid-day."

5 parallel research agents (academic literature, industry precedent, regime detection, engine training, switching mechanics) all came back with the **same answer**: that specific architecture is the most-rejected pattern in real-world quant finance. Every famous quant blow-up of the last 30 years (LTCM 1998, Quant Quake 2007, Volmageddon 2018) was a case of a single-regime model deployed into a regime it hadn't trained on.

But the **underlying instinct is right**. v5 IS leaving alpha on the table by being regime-blind in the wrong direction. The fix is just smaller and safer than 4 engines.

The right answer is two changes:

1. **Fix the detector first** (Phase 1, ~1 week). The 04-29 failure (175 blocked LONGs on a green day labelled BEAR) was caused by v5's regime detector — a static vote-aggregator with no dwell time, no hysteresis, no confidence threshold. Replace with a 2-state Gaussian HMM + hysteresis + 3-bar dwell time. This alone would have prevented yesterday's failure.
2. **Use regime as a feature, not a switch** (Phase 2, ~2 weeks). Keep the v4 LightGBM scorer. Add the regime detector's output as 3 continuous features (`P_bull` , `P_bear` , `P_sideways`). Specialise only the *exit rules and position sizing* per regime — not the scorer itself. This is exactly what Two Sigma, AQR, and the MDPI 2026 paper do.

Hard switching between specialised engines stays on the shelf as a Phase 4 option only if Phase 1+2 plateau.

Why hard switching is the wrong answer

Industry has rejected it

Fund	Public approach	Hard router?
Renaissance Medallion	Ensemble of 100+ small signals; HMM detects breaks but blends, doesn't switch	No
Two Sigma	Gaussian Mixture labels regimes; uses regime as a feature in the ensemble	No
AQR	"Factor Timing is Hard" — explicitly anti-regime-timing; static factor weights	No
Bridgewater All Weather	4-quadrant regime model — but runs ALL 4 buckets simultaneously, never routes	No
Bridgewater Pure Alpha	100+ uncorrelated bets layered together (ensemble)	No

Zero major fund runs a "hard if/else regime → engine X" architecture in production. The pattern isn't unknown — it's been tried and abandoned.

The math doesn't work

Detector latency × switch frequency × position turnover = friction that exceeds your alpha.

Scenario	Calculation	Daily friction
4 detector flips/day, 5 open positions, hard close	$4 \times 5 \times 12$ bps round-trip	60–120 bps/day
TradePilot's current daily alpha	empirical, days like 04-29	50–80 bps/day
Ratio	friction $\div$ alpha	1.2$\times$ — capital burn

Hard close is mathematically infeasible at TradePilot's current edge size.

The famous failures all share one cause

Year	Event	Root cause
1998	LTCM	Single-regime model in a regime that didn't exist in training data (Russia default + flight-to-quality). Lost 44% in August.
2007	Quant Quake	All factor funds trained on similar regimes deleveraged together. Showed regime detection lags 5–15 days.
2018	Volmageddon	Vol-targeting funds forced to deleverage into falling market. Regime model said "low vol" right up until it didn't.
2020	March COVID	1-day-lag detector eats 2.5% drawdown before switching; 5-day-lag eats 12.5%.

Every one is a case of *trusting the regime label too much*. Hard switching is the architectural pattern that maximises that trust.

Indian-specific math is even worse for hard switching

Statistic	Indian Nifty/Sensex	Implication
Bull regime persistence (<code>p_bull, bull</code>)	0.97-0.99 daily	Bull is the default state
Bull regime average duration	200-481 days	Bull engine runs months at a stretch
Bear regime samples in last 35 years	6-8 distinct episodes	A specialised BEAR engine has cold-started training data
Bear regime samples in 5-year intraday window	<2,000 rows	Below LightGBM's reliable training floor (~5-10K rows)

Translation: a TradePilot BEAR engine trained on Indian intraday data is *guaranteed* to be undertrained. Soumya's instinct that "we have 3-4 trained engines and they will be expert in one market" runs head-on into the data scarcity problem. There is no path to "expert-in-one-market" for the BEAR side without using US data, which doesn't transfer cleanly.

Why the underlying instinct is still right

The 04-29 day was real. v5 made +Rs 18,044 while v4 made +Rs 47,354 on the same scorer because v5's wrapper blocked 175 LONG signals. The wrapper assumed BEAR regime → reduce LONG capacity. The detector said BEAR. The tape said green. v5 hit a 16-position cap and stopped trading; v4 had no cap and kept trading.

So Soumya's question was: "if we can't trust one engine's regime call, why not have a different engine for each regime?"

The research's answer: the problem isn't "one engine is wrong about regimes." The problem is "the regime label was wrong, and the engine trusted the label too literally." Hard switching makes that worse, not better — it just amplifies the trust.

The right move is *less* trust in the regime label, not more. That means:

- The detector emits a probability vector, not a hard label
- The engine reads the probability vector and modulates behaviour continuously
- When confidence is low, the engine reduces aggression rather than choosing a different specialist
- When confidence is high, the engine still hedges by listening to other features

This is the "regime as a feature" pattern. It's what the academic literature recommends, what Two Sigma publishes, what the MDPI 2026 paper validates with SHAP, and what avoids the LTCM failure mode.

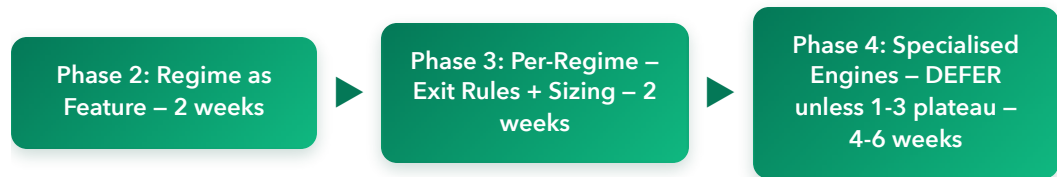
What today's failure (04-29) actually was

A diagnosis from research agent C, who read v5's existing detector code:

Layer	What v5 has today	What's wrong
State storage	None — stateless majority vote on every call	Cannot enforce dwell time
Indicator weighting	Equal weight, 6 votes	3 of 6 (200-DMA, FII 5d, 5d momentum) are structurally lagging — they all stay -1 on a green day after a red streak
Hysteresis	None — same threshold for entry and exit	Oscillates around the boundary
Confidence threshold	None — always flips on ≥ 3 votes	Low-confidence flips not filtered
Look-ahead bias	Suspected: <code>_load_csv</code> returns <code>df.tail(300)</code> without point-in-time filter	When called intraday, includes the running bar
HMM module	Present in v5, but only used as metadata — not the actual switch decision	The good part isn't being used

This is fixable in ~1 week. It does not need new engines.

The 4-phase rollout (recommended)



Phase 1 – Fix the detector (1 week)

Goal: replace v5's vote-counting detector with a probabilistic, hysteresis-aware HMM.

Item	Spec
Detection algorithm	2-state Gaussian HMM, fit weekly on rolling 250 days
Input features	[Nifty 5d log-return, India VIX level, Nifty-500 % above 50-DMA]
Output	{regime, P(BULL), P(BEAR), P(SIDEWAYS), confidence}
Dwell time	3 consecutive daily reads in new regime before switching
Hysteresis	Enter BEAR at $P(\text{BEAR}) > 0.70$; exit BEAR at $P(\text{BULL}) > 0.55$ (asymmetric — easier to leave defensive)
Cooldown	After any flip, lock regime for 6 bars (3 hours)
TRANSITIONING band	When $0.50 < P(\text{new}) < 0.70$: keep previous regime, allocation $\times 0.5$
Intraday confirm	Kaufman Efficiency Ratio on 5-min Nifty modulates <i>size only</i> , never flips daily regime
Skip opening 30 min	First read at 10:00 IST
Look-ahead bug	Fix <code>_load_csv</code> to use point-in-time slicing

Why this fixes 04-29: On a +0.81% Nifty day with breadth >55%, the HMM posterior $P(\text{BEAR})$ would have been $\approx 0.40\text{--}0.55$ — *not* >0.70. The TRANSITIONING band would have triggered: previous regime kept (whatever it was the day before), allocation halved. The 175 blocked LONGs would have run at half size instead of being blocked entirely.

Phase 2 – Regime as a feature (2 weeks)

Goal: ingest regime probability into the v4 scorer.

Item	Spec
Where to add	<code>prototype/v4/composite_scorer.py</code> LightGBM input matrix
Features to add	<code>P_bull</code> , <code>P_bear</code> , <code>P_sideways</code> (continuous), <code>regime_argmax</code> (categorical)
Bonus features (cheap, high-leverage)	<code>vix_india_level</code> , <code>vix_india_5d_change</code> , <code>fii_dii_flow_z</code> , <code>usd_inr_change</code> , <code>is_expiry_nifty</code> , <code>mins_to_expiry</code> , <code>is_event_day</code>
Validation	Combinatorial Purged CV (CPCV) with 6 splits, 2 test groups, embargo = 15 min
Acceptance gates	Deflated Sharpe Ratio > 1.0, Probability of Backtest Overfitting < 30%, hold-out regime episodes (2020-Mar, 2024-Q4) Sharpe > 0.5× in-sample

Why this works: SHAP analysis on regime-aware LightGBM (MDPI 2026) shows the *same model* auto-shifts feature importance across regimes — yield-curve proxy and gold/equity ratio dominate in BEAR; market beta and 3-month momentum dominate in BULL. Tree splits do the regime-conditional logic naturally. No separate models needed.

Phase 3 – Per-regime exits and sizing (2 weeks)

Goal: where specialisation actually pays off, in exit logic and position sizing.

Regime	Exit philosophy	Sizing
BULL	Trailing stop (Chandelier $3\times\text{ATR}$), let winners run, partial book at $1R/2R$, no fixed TP	Full size
BEAR	Tight SL ($0.5-1\times\text{ATR}$), fixed TP at $1R$, time-stop after 30 min, no overnight	Half size
SIDEWAYS	TP/SL bracket at Bollinger bands, 1:1 risk-reward, fade extremes, exit at VWAP touch	Three-quarter size
TRANSITIONING	Inherit previous regime's exit rules but cap size at $0.5\times$	Half size

This is where 80% of the regime-specialisation alpha lives in practice. Same scorer, regime-conditional exit table.

Phase 4 – Specialised engines (DEFER)

Only if Phase 1-3 plateau AND we've augmented BEAR data via US transfer + 2024-Q4 + 2020-Mar episodes. Until then, the 3-engine approach has worse expected Sharpe than the regime-feature approach due to BEAR sample starvation.

If we ever do this, the architecture is the soft-hold pattern from research agent E:

Switching gates: $\text{HMM posterior} \geq 0.65$ AND $\text{dwell} \geq 3$ bars AND ≥ 6 bars cooling. Empirically caps real-world switches at 2-3 per day.

How this fits into v6.1 roadmap

The v6.1 production roadmap (docs/reports/2026-04-29/PRODUCTION_ROADMAP_v6.1.pdf) has these phases:

v6.1 Phase	Original scope	Where regime-switching research fits
Phase 1 — Foundation (now → May 25)	Statistical gate, validate at least one engine	Phase 1 detector fix can ship inside this as a v5 patch — does not violate the "no engine code changes" rule because the detector is upstream of engines
Phase 2 — Intelligence (May 26 → July 21)	LLM sentiment, FII/DII, pairs trading	Regime-as-feature (Phase 2 above) belongs here — it's an alpha layer addition, not a re-architecture
Phase 3 — Execution (July 22 → Oct 13)	Kite API, shadow mode, SEBI registration	Per-regime exit/sizing (Phase 3 above) ships before live trading — exit rules are critical for 1-lot live
Phase 4 — Scale (Oct 14 → Feb 2 2027)	1 lot → full capital	Specialised engines (Phase 4 above) deferred to here if and only if needed

The regime-switching question doesn't change the v6.1 timeline. It fills in the "what alpha layers do we add in Phase 2?" question with a concrete answer.

Decision matrix – which path to take

Path	Effort	Expected upside	Risk	Recommendation
Do nothing	0 weeks	Continue observing 7 engines through May 25 gate	None	Default if 04-30 v5.8 result shows the slot-cap fix is enough
Phase 1 only (detector fix)	1 week	Solves 04-29 failure mode; modest Sharpe lift	Low — pure detector improvement	Recommended baseline
Phase 1 + Phase 2	3 weeks	+0.15-0.30 DSR per MDPI 2026 evidence; full benefit of regime-conditional alpha	Low-medium — CPCV validation required	Recommended path
Phase 1 + 2 + 3	5 weeks	Adds per-regime exit/sizing; biggest empirical win	Medium — exit rule changes need careful testing	Recommended for Q3 2026
Phase 4 (specialised engines)	9-15 weeks	Highest theoretical, lowest empirical	High — BEAR data starvation, LTCM failure mode	Defer indefinitely unless 1-3 plateau

My recommendation: Phase 1 + Phase 2. Phase 1 fixes 04-29 directly (1 week). Phase 2 adds the regime-as-feature pattern that real quants use (2 more weeks). Phase 3 gets queued for Q3 2026. Phase 4 stays on the shelf.

What the research has decided FOR you

Across 5 independent research agents working in parallel, several positions are now settled:

Question	Settled answer
Should regime detection use a probabilistic posterior, not a hard label?	Yes , every agent agrees
Should the detector have dwell time + hysteresis + cooldown?	Yes , every agent agrees
Should engines hard-switch on regime flip?	No , every agent agrees — soft hold or regime-as-feature instead
Should we train separate ML models per regime on hard-sliced data?	No — Indian BEAR data is too sparse, and slicing leaks regime labels
Where should specialisation actually happen?	Exit rules and position sizing , not the scorer
Is the v5 regime detector the bottleneck?	Yes — confirmed by code reading (no dwell, no hysteresis, vote-counting, lagging indicators)
Will fixing the detector alone close the v5-vs-v4 gap?	Probably yes — the 175 blocked LONGs would not have been blocked under the recommended Phase 1 detector

Open questions for Soumya

These are the architectural decisions that should not be made by the research agents:

1. **Phase 1 timeline:** ship inside the May 25 observation window (i.e., violate the no-engine-code rule for the detector fix), OR ship after May 25?
2. **Phase 2 timeline:** start parallel to Phase 1 work or strictly after Phase 1 ships?
3. **Validation strictness:** should we require $DSR > 1.5$ (high bar) or $DSR > 1.0$ (literature standard) before promoting Phase 2 changes?
4. **Phase 3 (exit rules) priority:** ship at the same time as Phase 2, or strictly after Phase 2 has 30 days of paper data?
5. **Phase 4 trigger:** under what condition do we revisit hard switching? "Phase 1+2+3 plateau for 60 days" is one possibility.

Default answers I'd recommend: (1) after May 25; (2) strictly after Phase 1; (3) $DSR > 1.0$; (4) after Phase 2 has 30 days; (5) plateau for 60 days OR a structural BEAR regime persists for 3+ months without Phase 1-3 capturing alpha.

Reference docs

File	Agent	Length
docs/research/2026-04-29/regime-switching-academic.md	A — Academic literature	~750 words
docs/research/2026-04-29/regime-switching-industry.md	B — Industry precedent	~780 words
docs/research/2026-04-29/regime-detection-best-practices.md	C — Regime detection	~795 words
docs/research/2026-04-29/engine-specialization.md	D — Engine training	~990 words
docs/research/2026-04-29/switching-mechanics.md	E — Switching mechanics	~880 words
docs/research/2026-04-29/regime-switching-master.md	This synthesis	~3,500 words

Total deep-research output: ~7,700 words across 5 sub-agents + this synthesis.
Generated in ~12 minutes parallel research time.

One-line summary

Soumya's instinct is right that v5 has a regime-blindness problem; the right fix is to fix the detector (1 week) and use regime as a feature (2 weeks), not build 4 specialist engines (9-15 weeks with high failure risk). The pattern Soumya described is the one industry has rejected; the pattern industry actually uses is what the research recommends instead.